



# Dr Charlay Wood, PhD

Plant Biochemist · Entrepreneur · Science Communicator

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## PROFILE

Plant biochemist working on how plants build new chemistry and how those pathways can be redesigned, with expertise in metabolism, enzyme discovery and natural product biosynthesis. Multi-award-winning presenter combining rigorous scientific thinking with strong communication, leadership and commercial skills. Experienced across academia, start-ups, direct sales and science outreach.

## EDUCATION

### PhD Biology

2020–2024

University of York, UK

- Thesis: The biosynthesis of *Securinega* alkaloids. Identified OLADO, a bacterial-like decarboxy-oxidase running the first committed step, published in *New Phytologist*.
- Skills: plant secondary metabolism, enzyme discovery, molecular biology, heterologous expression in *E. coli* and in planta, transcriptomics, metabolomics, chemical synthesis, isotope labelling, NMR, LC-MS.
- Supervisors: Dr Benjamin Lichman and Dr William Unsworth.

### MSc Biotechnology, Bioprocessing and Business Management

2016–2017

University of Warwick, UK

- Dissertation: Why are leaves green and not black? How can we increase photosynthetic efficiency?
- Supervisor: Prof. Lorenzo Frigerio.

### BSc (Hons) Biochemistry

2013–2016

University of Warwick, UK

- Final-year dissertation: Genes affected by *Hyaloperonospora arabidopsis* in *Arabidopsis thaliana*.
- First-year supervisor: Prof. Jim Beynon · Final-year supervisor: Prof. Miriam Gifford.

## RESEARCH EXPERIENCE

### Postdoctoral Researcher

2025–Present

University of Wisconsin–Madison, USA · PI: Prof. Hiroshi Maeda

- Investigating the shikimate pathway and aromatic amino acid biosynthesis.
- Increasing CO<sub>2</sub> fixation by up to 30% in crop plants, such as sorghum.
- Developing biochemical rerouting strategies to channel additional fixed carbon into high-value compounds such as pharmaceuticals and flavourings.

## ENTREPRENEURIAL & COMMERCIAL EXPERIENCE

### Invited Member, Business Development Working Group

2026–Present

American Society of Plant Biologists

- Invitation-only group meeting monthly on how the society builds sustainable revenue: which professional development programmes are worth running, who they are for, and what they are worth.
- The only early-career researcher on the group.

### Entrepreneurial Lead, NSF I-Corps

2025–Present

Natural Red Pigments, Madison, Wisconsin, USA

- An early-stage venture exploration of whether engineered soybean could supply a natural replacement for synthetic red colourants such as Red 40 (Allura Red AC). It is a commercialisation project, not an incorporated company.
- Led customer discovery through the NSF I-Corps programme, validating demand across food, beauty and plant processing.
- Leading the commercialisation thinking and contributing to the pigment biochemistry; the project was started by several of us together.

## Account Manager

2019–2020

*Alpha Gamma Solutions (Credico), London, UK*

- Face-to-face customer acquisition in direct sales and marketing, then leading my own sales and marketing team.
- Campaigns across utilities and telecommunications (broadband, gas and electricity) and charities including Marie Curie, Wood Green Animal Shelter, the National Deaf Children's Society and the Royal National Institute of Blind People.
- Campaign planning and execution, sales-team training, nationwide business trips and representing the company at corporate events.

## Entrepreneur in Residence

2017–2018

*Rocket Internet SE, Shanghai, China*

- Joined a post-launch start-up at a critical juncture; appointed Scrum Leader within one month.
- Given full operational control, managing five-figure investment budgets and all hiring decisions.
- Streamlined processes to reduce a six-person team to one employee supported by an outsourced team, with no loss of output.
- Mentored directly by the founder, a prominent serial entrepreneur.

## Founder, Danzarin

2016

*United Kingdom*

- First business, started during the business MSc. Bought experienced dancers' unworn Argentine tango shoes, restored them, and resold them to students who could not afford new pairs.
- Sold at student milongas across the UK, with pricing and demand tested directly on the customers.

## SCIENCE COMMUNICATION & OUTREACH

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### Research Highlights Editor

2026–Present

*The Plant Journal (Wiley) · continuing part-time editorial position*

- Selecting significant papers published in the journal and writing the Research Highlights that introduce them to a broad readership, including author interviews.

### Plantae Fellow

2026

*American Society of Plant Biologists · one-year fellowship*

- Themed weekly content, monthly webinars and collaborative science communication with fellows across three continents.

### Workshop Facilitator

2026

*Phytochemical Society of North America annual meeting, Madison, WI*

- Designed and facilitated "Seeds to Harvest: Discovery to Commercialisation", a 90-minute workshop and one of three run that year to get plant scientists thinking about how to translate their own ideas: is there a real problem, what is the solution, and who is the market.
- Invited a speaker who had built a company from plant natural products, and brought in colleagues from the university technology transfer office to work with participants.

### Conference Correspondent

2023 & 2025

*American Society of Plant Biologists*

- Covered two annual Plant Biology conferences as official ASPB Correspondent (Savannah, GA, 2023; Milwaukee, WI, 2025).

### Graduate Teaching Assistant

2020–2023

*University of York, UK*

- Supported undergraduate teaching in Biology: delivered lab sessions, assisted with tutorials, and guided students on practical and written work.

## ADDITIONAL EXPERIENCE

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### Argentine Tango Teacher & Dancer

2014–Present

*Various institutions (UK, China, USA)*

- Over a decade of teaching and performing across multiple cities, including self-run classes in Shanghai and ongoing involvement at UW–Madison.

### Retail Assistant (Seasonal)

2024

*Royal Botanic Gardens, Kew, UK*

- Christmas retail role at one of the world's most iconic botanical institutions, combining a passion for plants with visitor engagement.

## Online English Tutor

2020–2022

iTutor.com

- Delivered one-to-one English language lessons to learners aged 4–16 (plus adults), adapting materials across language and cultural barriers.

## English–Chinese Translator

October 2018

Heartbreak Theatre, International Arts Exchange, Wuxi, China

- Live translator between the English acting team and Chinese technical crew for a 10-day tour of *Romeo and Juliet*. Also translated the full script into Mandarin.

## Customer Service Assistant

2014–2015

ASDA, Leamington Spa, UK

- Part-time role during undergraduate studies. First point of contact for customer enquiries, responsible for resolving complaints before escalation. Cross-trained across multiple departments.

## PUBLICATIONS

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### Peer-reviewed research

- Wood, C. *et al.* (2026) Parallel evolution of plant alkaloid biosynthesis from bacterial-like decarboxylases. *New Phytologist*. DOI: 10.1111/nph.70884

### Research Highlights, written as Research Highlights Editor

- A clean slate for Rubisco engineering. *The Plant Journal* (Jul 2026). DOI: 10.1111/tpj.71043
- Naturally transgenic: circle-mediated HGT shapes the mitochondrial genome of *Mitrostemon yamamotoi*. *The Plant Journal* (Jun 2026). DOI: 10.1111/tpj.70989

### Selected science communication & other writing

- Four things wrong with how we teach plant science. *Plantae* (2026). Sole author.
- Beyond AlphaFold: the technologies waiting to change plant science. *Plantae* (Jun 2026). With Ruth Nichols and Sonia Balyan.
- The last piece of the nicotine puzzle. *ASPB Plant Science Research Weekly* (May 2026).
- The unwritten CV of a plant scientist. *Plantae* (Apr 2026). With Anastasia Kolesnikova.
- Quinine, deconstructed. *ASPB Plant Science Research Weekly* (Mar 2026).
- What vanilla, morphine and denim have in common. *Plantae* (Feb 2026). With the Plantae Fellows.
- How to build a start-up in plant sciences. *Plantae* (Feb 2026).
- Review: RUBY, a genetically encoded visible reporter for plant biology. *ASPB Plant Science Research Weekly* (Jan 2026).
- From Plant Biology 2023: the evolutionary origin of acutumene biosynthesis in Menispermaceae plants. *ASPB Plant Science Today* (Aug 2023). Interview with Prof. Jing-Ke Weng.
- Celebrating those who stand up for science against the odds. *The Biochemist* (Dec 2022). Interview with 2022 Maddox Prize winner Eucharía Nwaichi. DOI: 10.1042/bio\_2022\_140

## INVITED TALKS & WORKSHOPS

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- Phytochemical Society of North America — Madison, WI (Jun 2026). Workshop facilitator, “Seeds to Harvest: Discovery to Commercialisation” (90 minutes).
- University of Minnesota Duluth: invited seminar — Duluth, MN (Nov 2025). Parallel evolution of plant alkaloid biosynthesis from bacterial-like decarboxylases.
- Plant Chemical & Molecular Biology Seminar, UW–Madison — Madison, WI (Apr 2025). Biosynthesis of the *Securinega* alkaloids.
- University of York Postdoc Society: invited talk — York, UK (Jul 2024). How to get conference funding: a practical guide for early-career researchers.
- PhytoChem Talks: invited speaker — online (Dec 2023). First steps of  $\Delta$ 1-piperidine-derived alkaloid biosynthesis in *Flueggea suffruticosa*.
- University of York — York, UK (2022). Two-hour biology workshop designed and delivered for selected high school students, to encourage them into biology at degree level: slides, games and discussion points rather than a lecture.

## CONFERENCES

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- ASPB Plant Biology — Ottawa, Canada (Jul 2026). Attended as an invited member of the ASPB Business Development Working Group.
- International Conference on Machine Learning (ICML) — Seoul, South Korea (Jul 2026). Attended.
- New Phytologist Editor-in-Chief Symposium — Tartu, Estonia (Jul 2026). Speaker.
- Phytochemical Society of North America — Madison, WI (Jun 2026). **Workshop facilitator.**

- Hablemos de Ciencia — Madison, WI (Apr 2026). Poster, presented in Spanish.
- Biobased Symposium — Madison, WI (Mar 2026). **Speaker. Best Speaker Award.**
- NeurIPS — San Diego, CA (Dec 2025). Attended.
- ASPB Plant Biology — Milwaukee, WI (Aug 2025). **Conference Correspondent.**
- Phytochemical Society of North America — Toronto, Canada (Jun 2025). Poster.
- Botany University Challenge (BUC2024) — Oxford, UK (Aug 2024). **Quiz team member; runners-up.**
- High Value Biorenewables Network Annual Meeting — York, UK (2024). Attended.
- High Value Biorenewables Network Annual Meeting — Manchester, UK (Nov 2023). Attended.
- ASPB Plant Biology — Savannah, GA (Aug 2023). **Conference Correspondent.**
- Phytochemical Society of North America — East Lansing, MI (Jul 2023). **Speaker. Best Speaker Award.**
- Natural Products — Warwick, UK (Apr 2023). Speaker.
- RSC Chemical Biology and Bio-organic Group Symposium — Oxford, UK (Mar 2023). Poster.
- Chem@York — York, UK (Feb 2023). **Speaker. Best Speaker Award.**
- High Value Biorenewables Network Annual Meeting — Harrogate, UK (Nov 2022). Attended.
- Directing Biosynthesis VI — Edinburgh, UK (Jun 2022). Poster.
- Three Minute Thesis final, University of York — York, UK (May 2022). Speaker; finalist.
- North East Postgraduate Conference — online (Nov 2021). Poster.
- University of York Interdisciplinary Conference — York, UK (Sep 2021). Speaker.
- High Value Biorenewables Network Annual Meeting — Harrogate, UK (Mar 2021). Attended.
- Directing Biosynthesis V — online (Jun 2020). Attended.

## PROFESSIONAL QUALIFICATIONS

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- NSF I-Corps Entrepreneurial Lead
- Advanced R programming & data analysis (LC-MS, XCMS, metabolomics)
- Highest Commission Sales Representative (monthly award)
- Institutional biosafety & GMO handling certification
- Chemical safety training
- Responsible Conduct of Research (RCR) certification
- TEFL certified (120 hours)

## LANGUAGES

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English (native) · Mandarin Chinese (intermediate) · Spanish (intermediate)